DATA ANALYTICS AND MACHINE LEARNING FOR DIGITAL LOAN DISBURSEMENT ANALYSIS IN NBFCs

Field Project Report | B.Sc. Data Science | Academic Year 2026–2027

Student Name: _____

Roll No.: _____

Division: _____

Internal Guide: _____

College: Atharva College of Hotel Management & Catering Technology

CERTIFICATE

This is to certify that _____, student of B.Sc. Data Science, has completed the field project titled “Data Analytics and Machine Learning for Digital Loan Disbursement Analysis in NBFCs” during the academic year 2026–2027 under the guidance of the undersigned. The project follows the prescribed academic format.

Internal Guide: _____ Course Coordinator: _____ Principal: _____

STUDENT DECLARATION

I, _____, student of B.Sc. Data Science, declare that this project report titled “Data Analytics and Machine Learning for Digital Loan Disbursement Analysis in NBFCs” is my academic project work. The analysis is based mainly on public secondary data and Python-based analysis. Sources used in the project are acknowledged in the references.

Student Signature: _____ Date: _____

ACKNOWLEDGEMENT

I would like to thank my project guide for the guidance and feedback provided during this project. I am also thankful to the college and the Department of Data Science for giving me the opportunity to work on a practical data-science project. I thank the Reserve Bank of India for making public statistical information available, which made this secondary-data study possible. Finally, I thank everyone who supported me during the project.

ABSTRACT

This project studies digital loan disbursement patterns in Non-Banking Financial Companies (NBFCs) using public data from the Reserve Bank of India (RBI). The project uses Python for data cleaning, analysis and visualisation, and machine-learning methods to study observable lending patterns. The dataset contains 305 NBFC-period observations from the RBI digital-lending data used in this project. The analysis covers digital disbursement amount, number of loans and average ticket size.

The machine-learning stage uses 135 complete NBFC-to-next-period transitions from 51 NBFC groups. Classification models are used to predict whether the next observed period has a higher disbursement amount. Regression models are used to estimate the next observed disbursement amount. K-Means is used to group NBFCs by lending behaviour, while Isolation Forest is used to flag potentially unusual observations.

The best classification model by F1 score in the grouped cross-validation used here is Random Forest Classifier, with an accuracy of 73.33% and F1 score of 80.85%. The regression model with the lowest RMSE is Gradient Boosting Regressor, with an RMSE of ₹1112.69 crore. These results show how machine learning can be applied to observable lending data, but they do not prove that internal NBFC analytics systems caused lending changes.

Keywords: NBFC, digital lending, data analytics, machine learning, Python, loan disbursement, classification, regression, clustering, anomaly detection

TABLE OF CONTENTS

Chapter 1 – Introduction

Chapter 2 – Literature Review

Gap Analysis and Objectives

Chapter 3 – Research Methodology

Chapter 4 – Data Analysis, Visualisation and Machine Learning

Chapter 5 – Findings, Conclusion and Recommendations

References

Appendices

CHAPTER 1: INTRODUCTION

1.1 Background

NBFCs are financial institutions that provide credit and other financial services but are different from traditional banks. Digital lending has increased the use of technology in loan applications, processing, monitoring and disbursement. Digital processes create large amounts of structured data that can be studied using data analytics.

In this project, the main focus is not on a private NBFC's internal software. Instead, the project uses publicly available RBI digital-lending data to understand observable patterns in loan disbursement, loan count and average ticket size.

1.2 Data Analytics in NBFC Lending

Data analytics means collecting, cleaning and studying data to find useful patterns. In lending, analytics can help an organisation understand loan volumes, product mix, changes over time and unusual activity. Good analytics depends on good data. If data is incomplete or inconsistent, the final results may also be unreliable.

1.3 Role of Machine Learning

Machine learning is a part of data science where a computer learns patterns from existing data and uses those patterns to make predictions or group similar observations. In this project, machine learning is used only for the variables available in the public dataset.

1.4 Problem Statement

Digital lending creates a large amount of lending information, but raw tables are difficult to understand directly. There is a need to clean the data, visualise important patterns and use suitable machine-learning techniques to study future observable lending behaviour, group NBFCs and identify unusual observations.

1.5 Objectives

- To collect and understand public RBI digital-lending data related to NBFCs.
- To clean and organise the data using Python and Pandas.
- To visualise digital disbursement, loan count and average ticket size using Python.
- To study the relationship between loan count and disbursement amount.
- To build a classification model for predicting whether the next observed disbursement increases.
- To build a regression model for estimating the next observed disbursement amount.
- To group NBFCs using K-Means clustering based on observable lending behaviour.
- To identify potentially unusual lending observations using Isolation Forest.
- To interpret the results in simple business terms and state the limitations of the available data.

1.6 Scope

The project is limited to the public RBI data used in the cleaned project dataset. It does not include borrower-level personal information, credit scores, income, approval or rejection records, repayment/default records, or private NBFC analytics-system logs.

1.7 Significance

The project demonstrates how a data-science workflow can turn a public financial dataset into useful charts, statistical measures and machine-learning outputs. It also gives practical experience with data cleaning, feature engineering, model evaluation and interpretation.

1.8 Structure of the Report

Chapter 2 explains the main ideas from existing literature. Chapter 3 explains the data and methods. Chapter 4 presents Python analysis and machine-learning results. Chapter 5 gives findings, conclusion and recommendations.

CHAPTER 2: LITERATURE REVIEW

Digital lending connects financial services with digital platforms and data-driven processes. Research and regulatory work on digital lending commonly discusses customer experience, responsible lending, data use, privacy, risk management and technology-enabled credit processes.

Machine learning is widely used in financial data analysis for classification, prediction, segmentation and anomaly detection. Classification models can separate observations into categories, regression models estimate numerical values, clustering groups similar observations, and anomaly detection highlights observations that look different from the majority.

A major theme across data-science literature is that model quality depends on the quality of the input data. Data must be cleaned, checked and transformed before a model is trained. Model results should also be evaluated using appropriate test procedures rather than only looking at training performance.

2.1 Research Gap

The available public RBI digital-lending data provides useful aggregate information about digital loan amounts and loan counts, but it does not provide borrower-level information or private NBFC analytics-system data. This creates a practical gap: a student project can analyse observable lending patterns, but it cannot directly test how a particular NBFC's private analytics system changes individual credit decisions.

This project addresses the practical data-science side of the gap by combining public financial data, Python visualisation and multiple machine-learning techniques in one workflow. The focus is therefore on demonstrating how available data can be analysed and modelled, while clearly stating what the data cannot prove.

2.2 Objectives

The objectives listed in Chapter 1 guide the analysis and model design in Chapter 4.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Data Source

The main source is public digital-lending information published by the Reserve Bank of India. The cleaned working dataset contains 305 NBFC-period rows representing 61 visible respondent IDs across five reporting periods in the extracted data.

3.2 Dataset Variables

Variable	Meaning
NBFC respondent ID	Anonymous respondent identifier used in the extracted dataset
Period	Reporting period
Digital disbursement (₹ crore)	Observed digital loan amount disbursed
Digital loan count	Number of digital loans
Average ticket size (₹)	Average disbursement amount per loan

3.3 Data Preparation

- Read the cleaned RBI-derived CSV into Pandas.
- Converted numeric columns into numeric format.
- Removed duplicate NBFC-period combinations.
- Kept the reporting periods in their correct order.
- Created average ticket size from amount and loan count where required.
- Created transition records from one observed period to the next for machine learning.
- Used only complete transitions for the ML stage.

3.4 ML Dataset

The transition dataset contains 135 complete transitions across 51 NBFC groups. Each transition uses current-period observable lending variables to study the next observed period.

3.5 Machine Learning Methods

Method	Simple meaning	Purpose
Logistic Regression	Predicts a yes/no outcome	Baseline classification
Random Forest Classifier	Uses many decision trees	Predict increase/decrease
Ridge Regression	Linear numerical prediction with regularisation	Baseline regression
Random Forest Regressor	Many trees predict a number	Disbursement prediction
Gradient Boosting Regressor	Builds trees step by step to reduce errors	Disbursement prediction
K-Means	Groups similar observations	NBFC segmentation
Isolation Forest	Finds observations that look unusual	Anomaly detection

3.6 Model Evaluation

Classification was evaluated using accuracy, precision, recall and F1 score. Regression was evaluated using MAE, RMSE and R^2. A 5-fold GroupKFold procedure was used for classification and regression, with NBFC respondent ID used as the group. This reduces the risk of placing observations from the same NBFC in both training and testing folds.

3.7 Research Limitations

- The public data is aggregate rather than borrower-level.
- The dataset does not directly measure each NBFC's internal analytics system.
- The number of time periods is limited.
- The ML models therefore learn observable lending patterns, not private credit-decision rules.
- Model results should not be treated as causal proof.

CHAPTER 4: DATA ANALYSIS, VISUALISATION AND MACHINE LEARNING

4.1 Dataset Overview

The final cleaned dataset contains 305 NBFC-period observations. It represents 61 visible respondent IDs across five reporting periods. The ML transition dataset contains 135 complete current-to-next-period transitions.

Period	Disbursement observations (₹ crore)	Loan-count observations	Total disbursement (₹ crore)	Total loans	Weighted average ticket (₹)
FY2017	18	20	741	101,588	72,941.69

FY2018	29	31	3,539	1,147,545	30,839.75
FY2019	37	38	13,256	8,407,168	15,767.50
FY2020	53	53	42,297	49,211,386	8,594.96
As on Dec 31 2020	59	59	22,780	22,389,181	10,174.56

4.2 Python Visualisation

Python was used to create the main charts. The charts are included in the project package and can be recreated from the cleaned CSV.

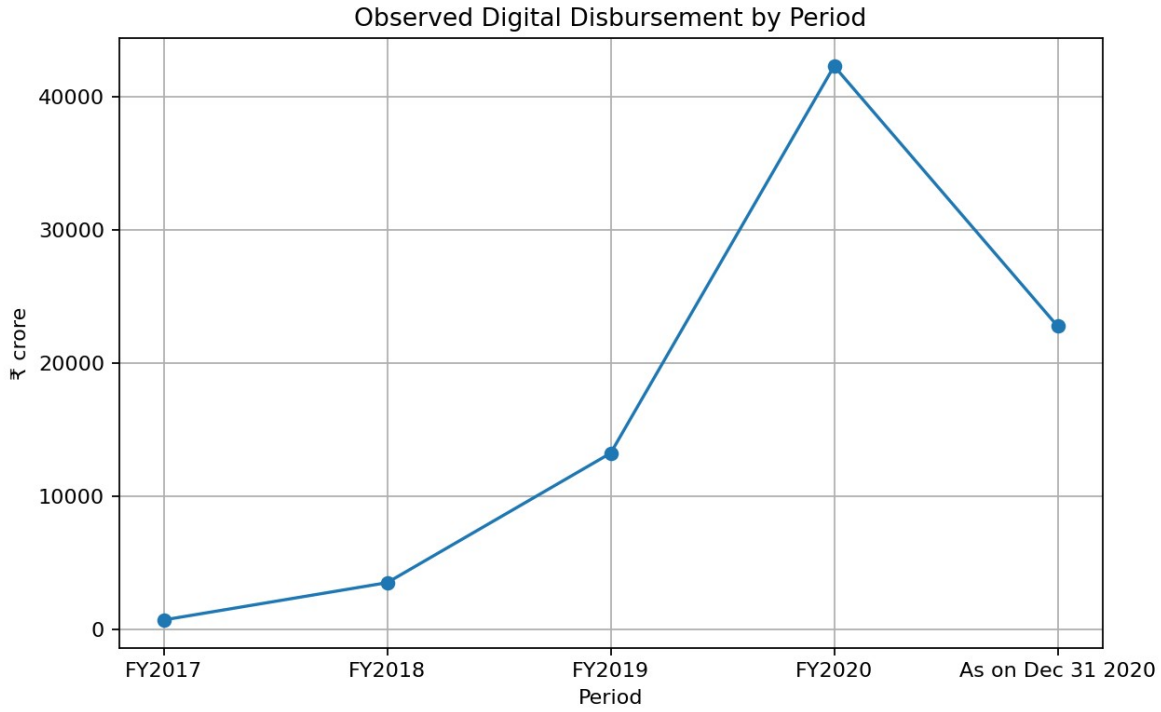


Figure 4.1: Observed digital disbursement trend.

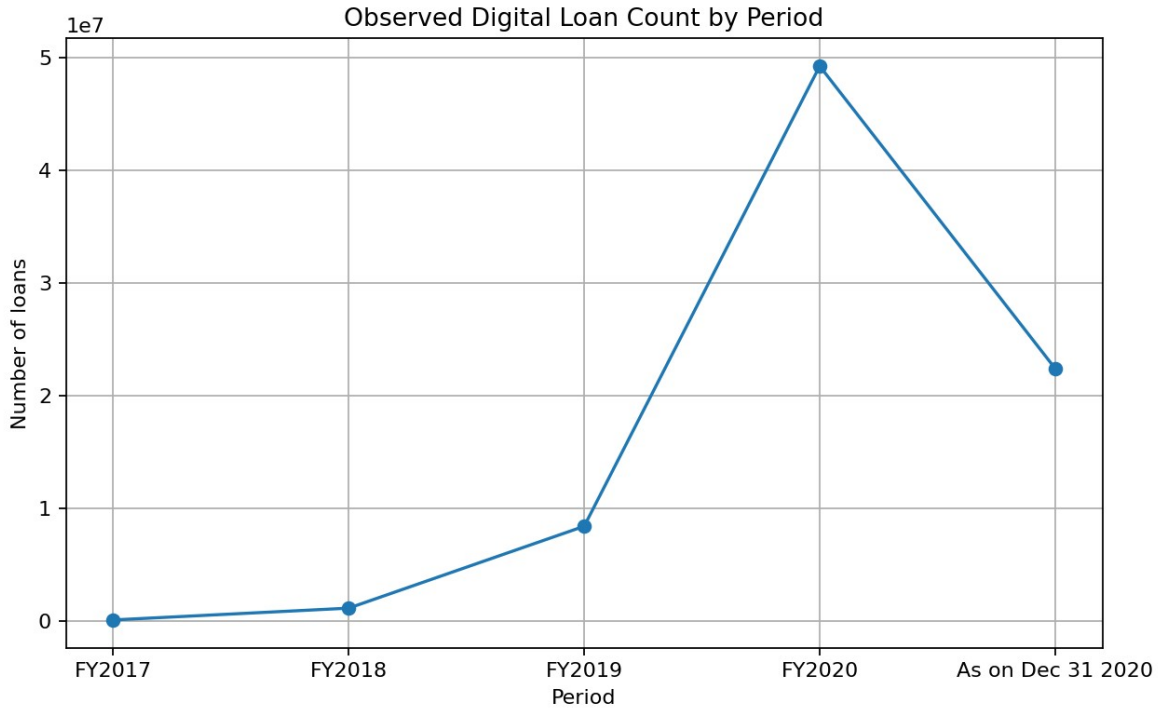


Figure 4.2: Observed digital loan count trend.

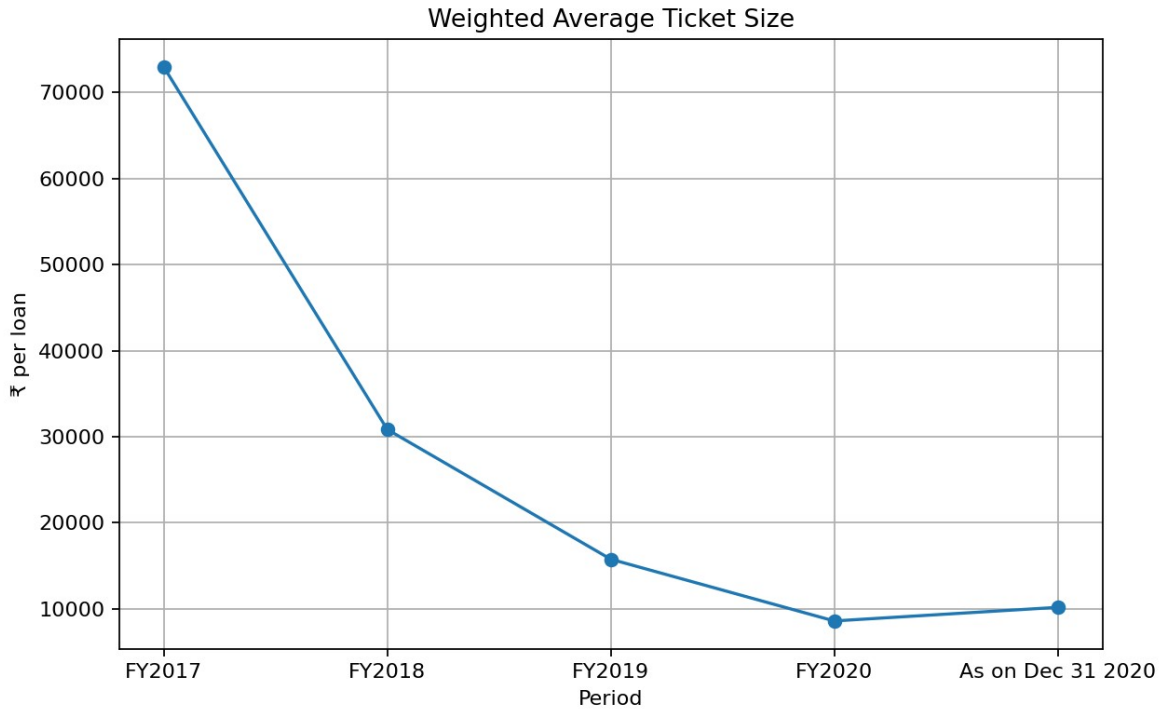
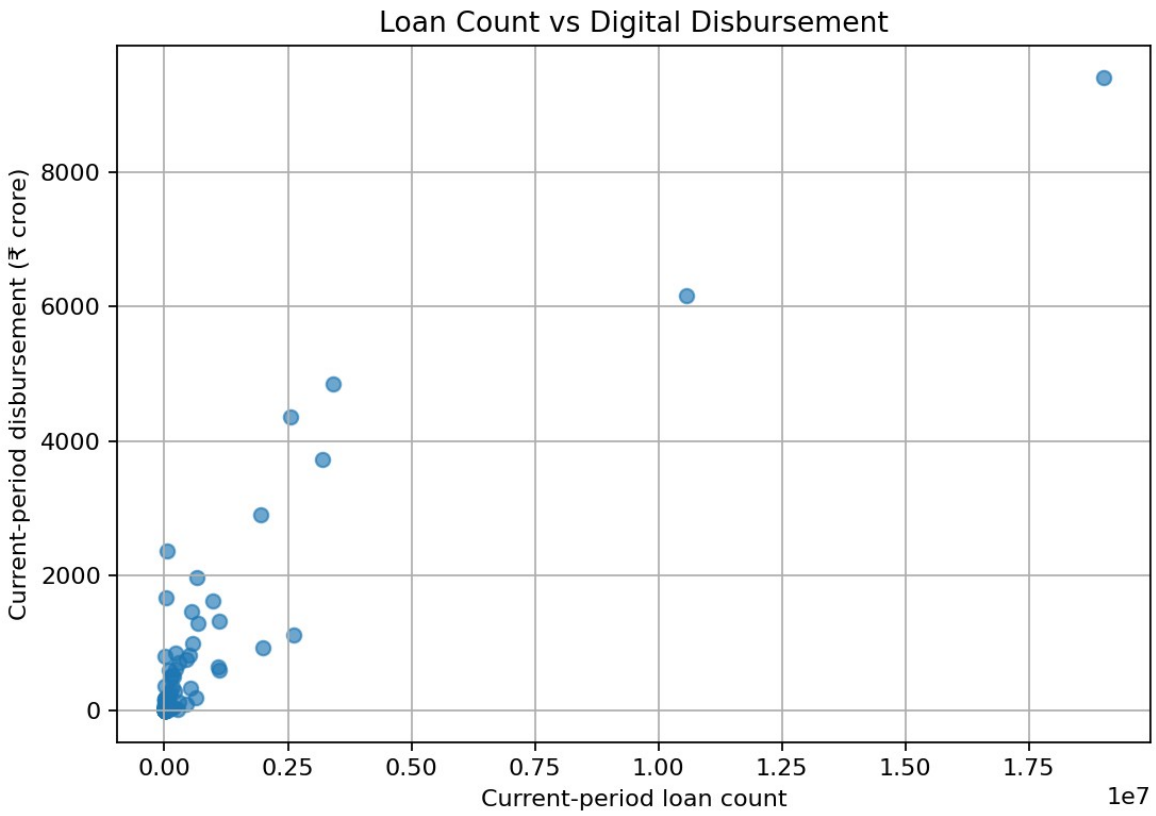


Figure 4.3: Weighted average ticket size.

4.3 Loan Count and Disbursement Relationship

In the transition data, the Pearson correlation between current loan count and current digital disbursement is approximately 0.893. This indicates a strong positive association in the available observations. Correlation shows association, not causation.

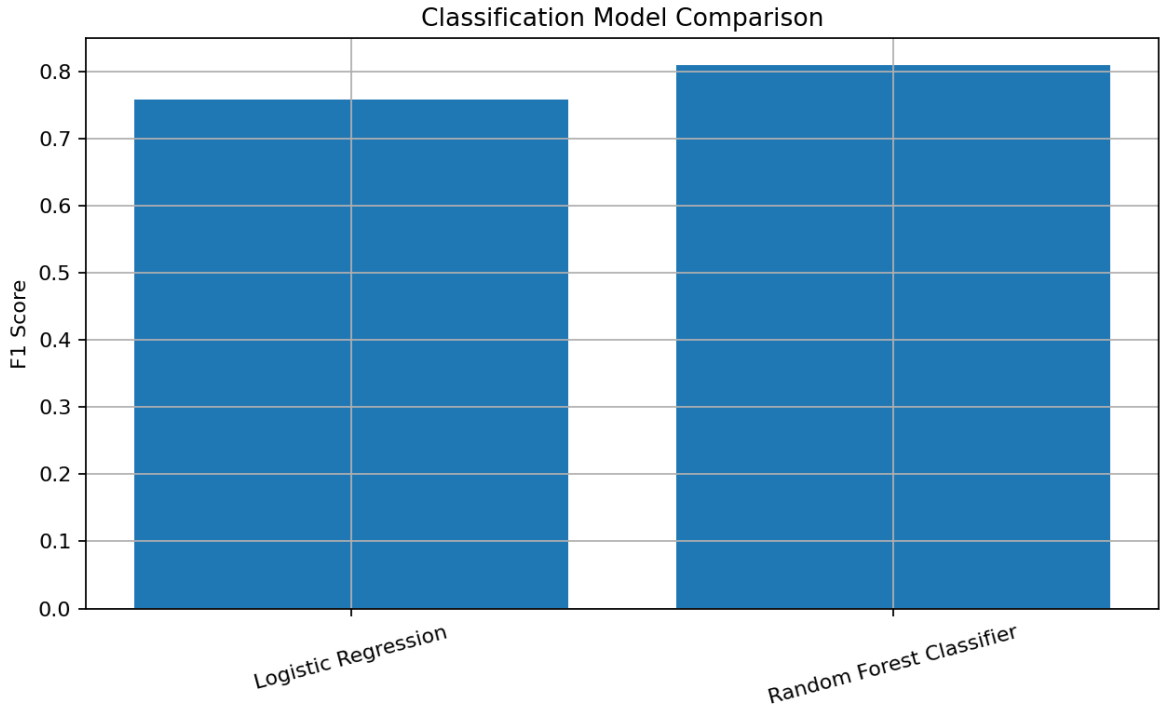


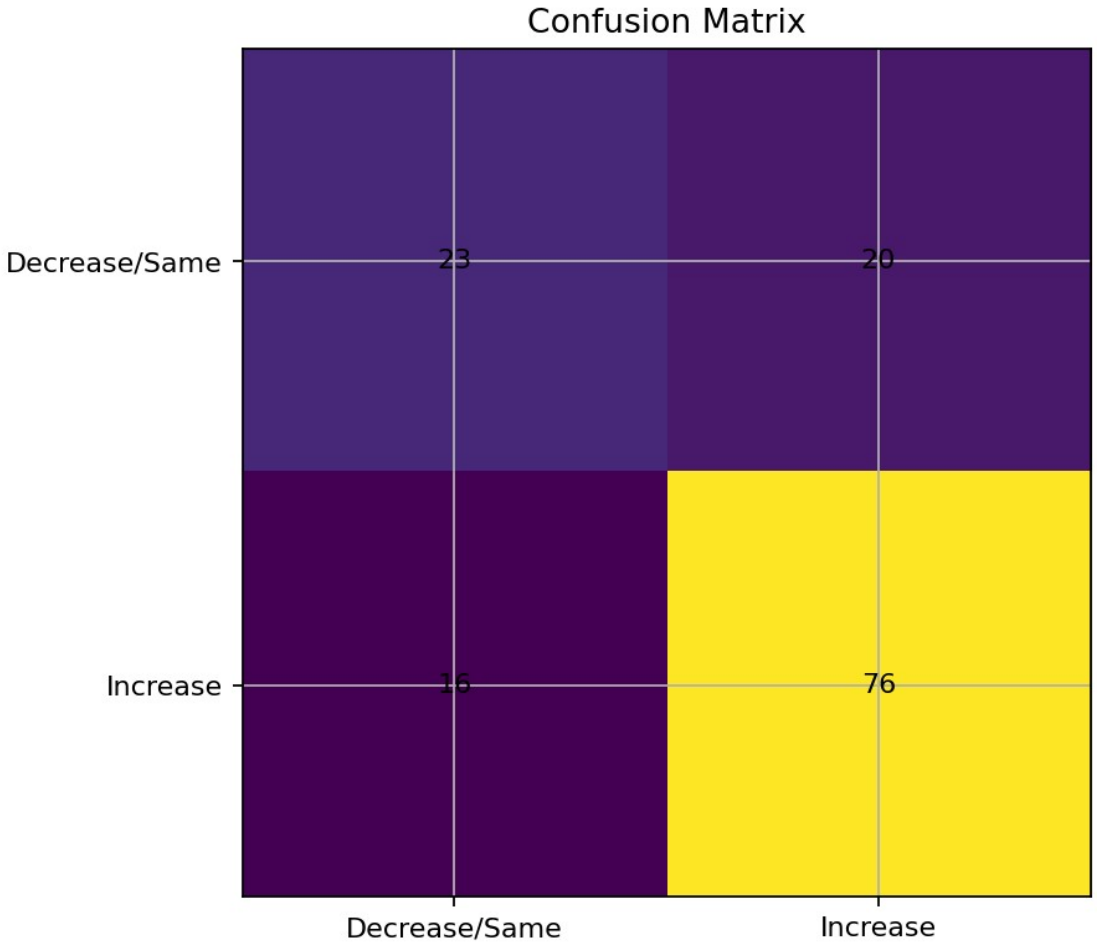
4.4 Classification Model

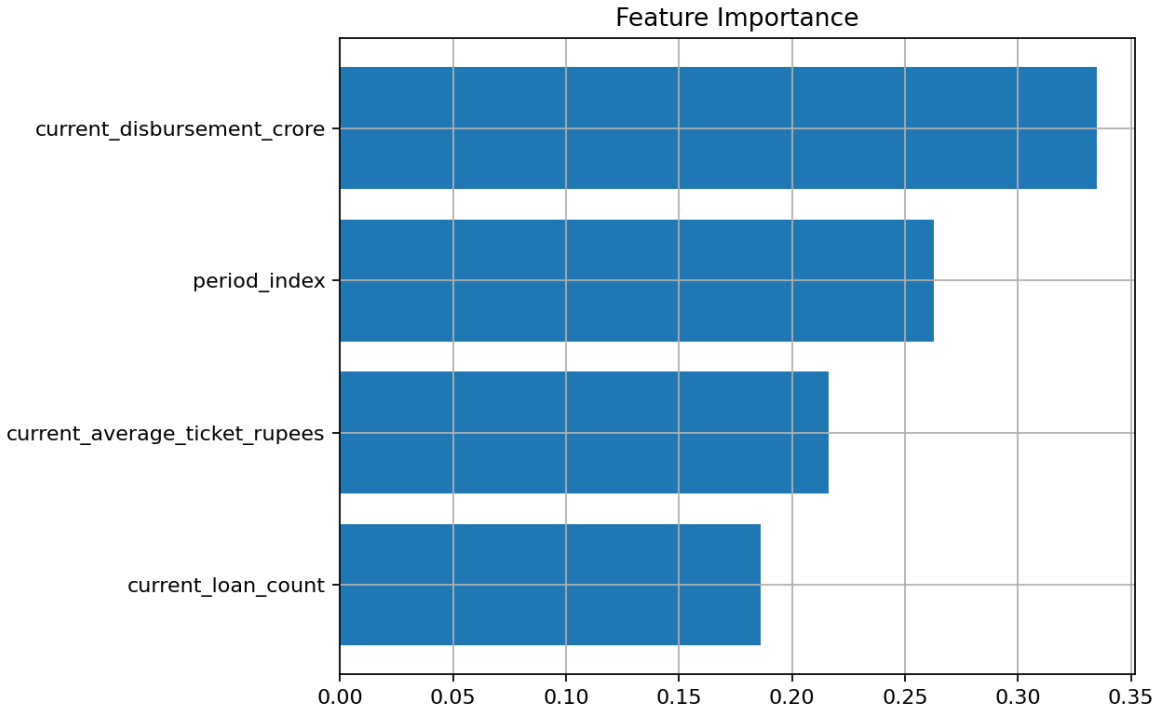
The classification task asks: “Will the next observed period's digital disbursement be higher than the current period?” The model uses current disbursement, current loan count, current average ticket size and period index.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.696	0.831	0.696	0.757
Random Forest Classifier	0.733	0.792	0.826	0.809

The highest F1 score in this evaluation is from Random Forest Classifier. Its F1 score is 0.809.


Classification Model Comparison
F1 Score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Logistic Regression
Random Forest Classifier


Confusion Matrix
Decrease/Same
Increase
23
20
16
76
Decrease/Same
Increase

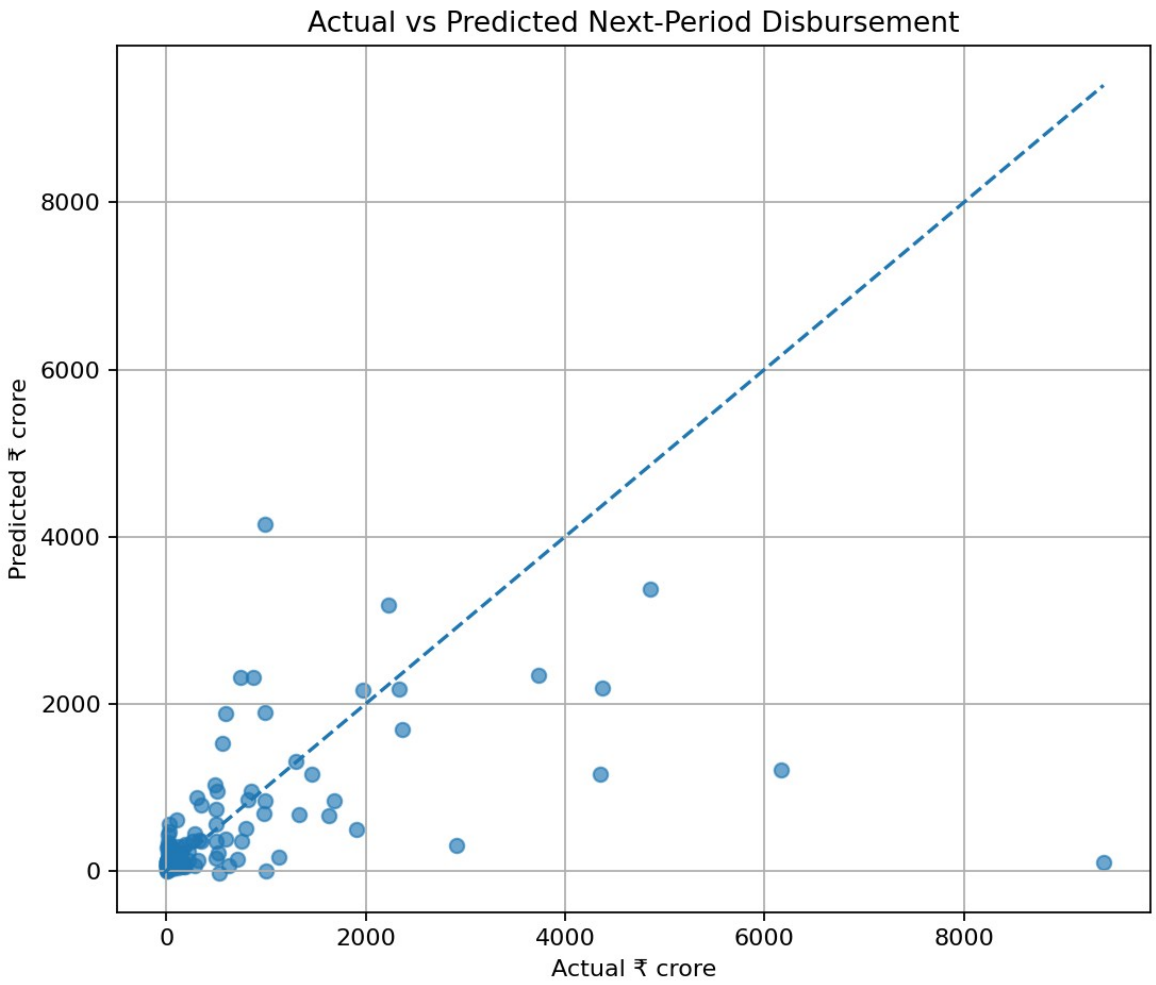


4.5 Regression Model

The regression task estimates the next observed period's digital disbursement amount in ₹ crore.

Model	MAE (₹ crore)	RMSE (₹ crore)	R^2
Gradient Boosting Regressor	433.95	1112.69	0.221
Random Forest Regressor	526.39	1165.77	0.145
Dummy Median	569.39	1357.90	-0.161
Ridge	733.20	1571.10	-0.554

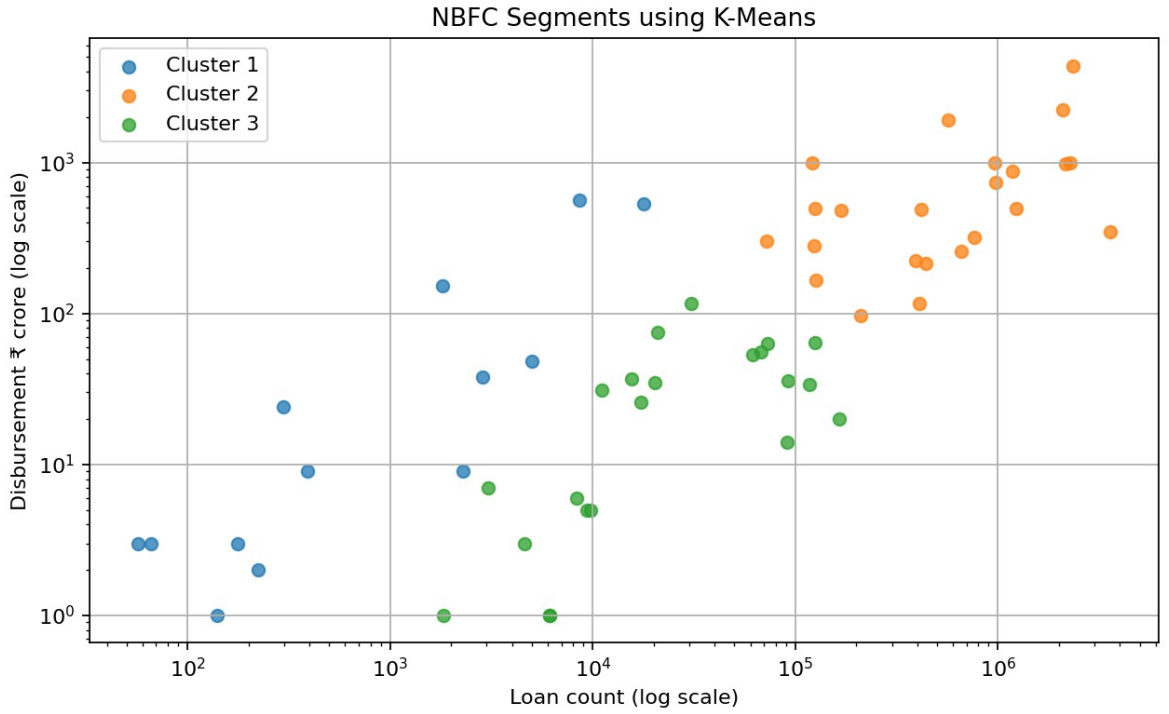
The lowest RMSE in this evaluation is produced by Gradient Boosting Regressor. Its RMSE is ₹1112.69 crore and R^2 is 0.221.



4.6 K-Means Clustering

K-Means was used on the latest observed period. The model used log-transformed disbursement, loan count and average ticket size so that very large values would not dominate the grouping.

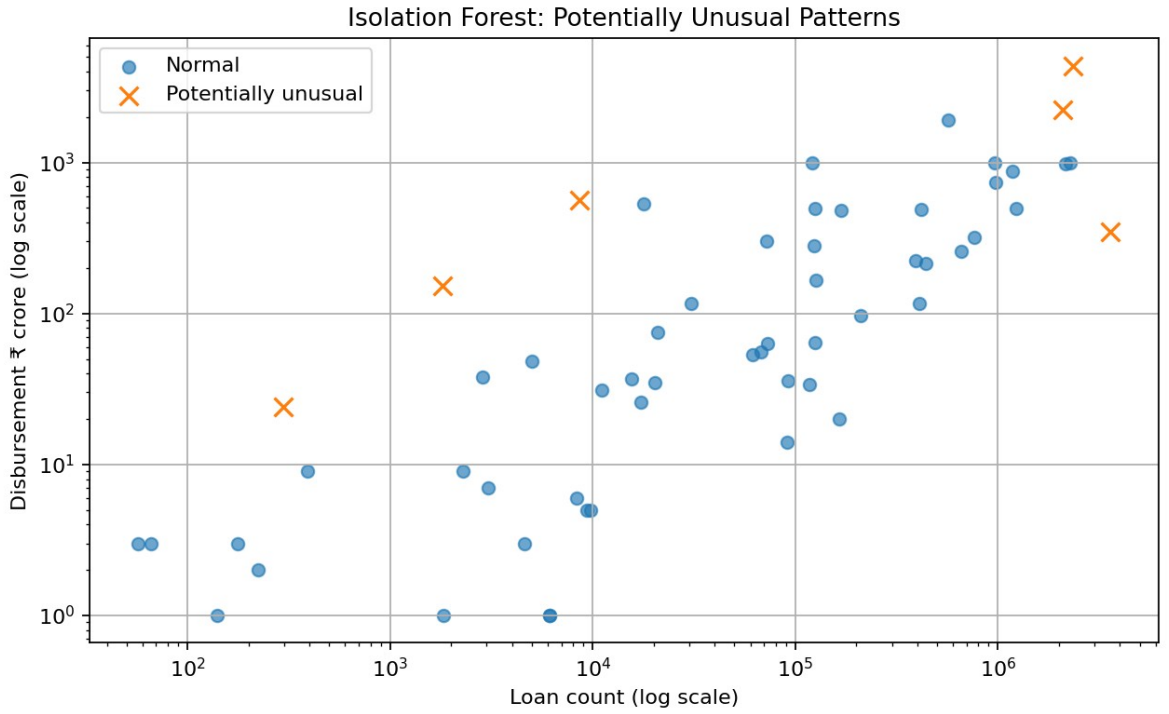
Cluster	NBFC count	Mean disbursement (₹ crore)	Mean loan count	Mean ticket (₹)
1	13	106.38	3,051	339,102.90
2	23	798.70	930,051	16,016.33
3	22	31.36	43,559	11,552.60



The clusters should be described as data-driven groups, not as official RBI categories.

4.7 Isolation Forest

Isolation Forest flagged 6 of 58 complete latest-period observations as potentially unusual. These observations are candidates for further checking; an anomaly flag does not automatically mean an error or fraud.



4.8 Overall Interpretation

The analysis shows that digital lending data can be used for several data-science tasks. Descriptive charts show changes in amount, loan count and ticket size. Classification gives a structured way to predict direction of change. Regression provides a numerical estimate. Clustering reveals groups of similar lending behaviour, and anomaly detection identifies observations that deserve closer inspection.

At the same time, the models have limits. The regression R^2 is not high, so the model does not explain all variation in the next-period amount. This is expected because many factors that influence lending are not present in the public dataset. The results should therefore be used as an analytical demonstration, not as a production lending system.

CHAPTER 5: FINDINGS, CONCLUSION AND RECOMMENDATIONS

5.1 Findings

- The cleaned dataset contains 305 NBFC-period observations from 61 visible respondent IDs.
- The ML stage uses 135 complete transitions across 51 NBFC groups.
- The classification models can identify the direction of the next observed disbursement with useful but imperfect performance; the best F1 score is 0.809.
- The best regression model by RMSE has an RMSE of ₹1112.69 crore and R^2 of 0.221.
- K-Means produced three data-driven NBFC groups in the latest complete period.
- Isolation Forest flagged 6 latest-period observations for further inspection.
- Python visualisation made the main lending patterns easier to understand than raw tables alone.
- Because the dataset is aggregate, the models do not directly measure borrower-level credit decisions or private analytics systems.

5.2 Conclusion

This project shows a complete data-science workflow for studying digital loan disbursement in NBFCs. The workflow starts with public financial data, continues through cleaning and visualisation, and then applies machine learning for prediction, segmentation and anomaly detection.

The main learning from the project is that machine learning is useful only when the data and the question match. The available RBI data is suitable for studying observable lending patterns, but it is not sufficient to build a borrower-level credit approval or default model. The project therefore keeps its claims within what the data can support.

5.3 Recommendations

- NBFC analytics teams should maintain clean, consistent and well-defined lending data.
- Prediction models should be tested on unseen data and monitored over time before any real business use.
- Unusual observations should be investigated by domain experts instead of being treated automatically as errors or fraud.
- Future studies should combine public aggregate data with approved, anonymised borrower-level datasets where legally and ethically possible.
- Future work can include more time periods, product-level variables and operational variables such as processing time, approval rate and repayment performance.

5.4 Future Scope

A future version can use a larger time-series dataset, product-level data and anonymised borrower-level variables. With suitable data, the project could explore default prediction, approval probability, loan

demand forecasting and explainable machine learning. Such models would require careful privacy, fairness and validation controls.

REFERENCES

1. Reserve Bank of India. Report of the Working Group on Digital Lending including Lending through Online Platforms and Mobile Apps. RBI.
2. Reserve Bank of India. Sectoral Deployment of Bank Credit by NBFCs. RBI Database.
3. Reserve Bank of India. RBI Digital Payments Index. RBI.
4. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
5. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.
6. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9(3), 90–95.

Official RBI digital lending report page: https://www.rbi.org.in/Scripts/PublicationReportDetails.aspx?UrlPage=&ID=1189

Official RBI NBFC sectoral credit page: https://www.rbi.org.in/Scripts/Data_Sectoral_Deployment_NBFC.aspx

Official RBI statistics portal: https://statistics.rbi.org.in/

APPENDICES

Appendix A – Field Study Evidence

This project is based on public secondary data. No primary field survey is claimed in the analysis. If the college requires a physical field-study photograph or signed evidence, attach genuine evidence separately rather than creating artificial evidence.

Appendix B – Google Form

Not used for the main analysis because the project is designed as a secondary-data and machine-learning study. If the college specifically requires a Google Form, obtain guide approval before adding one.

Appendix C – Area Map

Not applicable to the secondary-data analysis. Add a genuine project-relevant map only if required by the college.

Appendix D – Google Form Responses

Not applicable because no Google Form responses are used in the main analysis.

Appendix E – Raw Data

The raw/cleaned working CSV and all ML datasets are included in the project ZIP under 03_Data. The Python scripts and saved models are included under 02_Python_ML.

Important: guide-interaction signatures, plagiarism percentage, student details and institutional signatures must be completed by the student/college. They have not been fabricated in this package.